

Introduction to R -modules

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Abstract

These are notes for a talk given to the [Society of Mathematics](#) in the Fall 2026 semester, as part of their semesterly Colloquium. This semester's topic is on symmetries and invariants. In section 1, we will motivate modules by discussing how the structure of an abelian group is precisely that of a $\mathbb{Z}$ -module. In section 2, we will introduce rings, and modules over a ring. In section 3, we introduce standard linear algebra constructions, such as R -linear maps, kernels, images, and products. We will also discuss how a nontrivial torsion submodule obstructs the existence of a basis for a module over an arbitrary ring, which will motivate quotients as a way of correcting this issue. We will conclude in section 4 with an introduction to chain complexes and homology, since chain complexes are a powerful modern tool for producing computable and interesting invariants from abstract objects.

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1 Motivation

So far in the Colloquium, we have seen the definition of a group, and in particular, an *abelian group*. We have focused on groups as symmetries of an object, but they are still interesting algebraic structure to study abstractly. We will begin by investigating abelian groups, since they are generally easier to investigate.

Definition 1.1. A *group* is a set G and an operation $\cdot : G \times G \rightarrow G$ called *the group operation* or *group multiplication*, which satisfies the following properties, called the *group axioms* (here we often write $a \cdot b$ as ab and call this their *product*):

1. For all $a, b, c \in G$, we have $(ab)c = a(bc)$. This is called *associativity*.
2. There exists some element $e \in G$ called the *identity*, with the property that for all elements $a \in G$, we have $ea = ae = a$.
3. For all $a \in G$, there is an element a^{-1} called the *inverse* of a , with the property that $aa^{-1} = a^{-1}a = e$.

If G also has the property that for all $a, b \in G$, we have $ab = ba$, then G is called an *abelian group*. This property is called the *commutative property*.

Note that the requirement that $\cdot$ be a function $G \times G \rightarrow G$ is the same as the *closure* axiom that many books use. This simply means that if $g, h \in G$, then $gh \in G$.

The set G is called the *underlying set* of the group G . We may sometimes refer to a group as a pair $(G, \cdot)$, where G is the underlying set and $\cdot$ is the group operation. This is mainly for notational convenience.

Note that in the above definition, we call the group operation “*multiplication*”, and use common notation for multiplication when writing it. However, there are groups where it would make more sense to refer to the group operation as something else. For example, consider the group of integers numbers $\mathbb{Z}$ under addition. This is indeed a group, since addition is associative, $0 \in \mathbb{Z}$ is the identity, and if $n \in \mathbb{Z}$, there is an additive identity $-n \in \mathbb{Z}$. It would make no sense to use the notation in definition 1.1 here, since I would then be writing $n \cdot m$ to mean $n + m$. I should not have to explain why this is a terrible idea.

Thus, there are two main notational conventions for groups. There is *multiplicative notation*, where the group operation is denoted by multiplication, and there is *additive notation*, where the group operation is denoted by addition. It will be useful for us to adopt the following convention.

Convention 1.2. If G is a group, then unless otherwise stated, we will use multiplicative notation if it is *not* an abelian group. If G is abelian, we will use additive notation (unless otherwise stated). To simplify this, I will often write $(G, +)$ to denote that we are using additive notation, and $(G, \cdot)$ to denote that we are using multiplicative notation.

Notation 1.3 (Additive notation). Let $(G, +)$ be a group using additive notation. Then for $a, b \in G$, the group operation is written as $a + b$. The inverse of a is written as $-a$ and is called *negative a*. Finally, the identity element of G is written as $0 \in G$, and is also called *zero*. We may occasionally write 0_G instead of 0 to specify that 0_G is an element of G . This is useful when dealing with multiple groups at once.

We want to study groups abstractly, and abelian groups are a much simpler type of group due to their commutative property. As such, we will begin by investigating some basic properties of abelian groups.

Let’s start with a basic observation. Suppose that $(G, +)$ is a group, and $a \in G$ is some element. Then if $n \in \mathbb{Z}_{>0}$ is a *positive* integer, then I can define the following element:

$$n \cdot a := \underbrace{a + a + \cdots + a + a}_{n \text{ times}}.$$

This is very sensible notation; multiplication of integers $n \cdot m$ is just repeated addition $m + m + \cdots + m$, so we simply use the same notation for repeated addition in groups. This allows us to “multiply” a positive integer n with an element $a \in G$ of an abelian group.

We can extend this notation to all integers. First, recall that if n is an integer, then $0 \cdot n$ is simply 0 . This suggests that 0-fold repeated addition should be the identity element. Thus, if $a \in G$, we define

$$0 \cdot a := 0_G.$$

The 0 appearing on the left is the integer 0, and the 0 appearing on the right is the identity of the group G . Note that calling our group’s identity “0” plays very nicely with this “multiplication by $\mathbb{Z}$ ” notation.

Finally, we define multiplication by -1 to simply be

$$-1 \cdot a = -a.$$

We can then define

$$-n \cdot a := (-1 \cdot n) \cdot a = -1 \cdot (n \cdot a) = -(n \cdot a).$$

This allows us to define a multiplication operation $\mathbb{Z} \times G \rightarrow G$. We call this a “ $\mathbb{Z}$ -action” on the group G . we will often drop the multiplication dot, and will simply write na instead of $n \cdot a$.

Remark 1.4. This works in non-abelian groups, is written as a^n , and is called exponentiation.

We can now observe some basic, but extremely important, properties of abelian groups.

Proposition 1.5. Let $(G, +)$ be an abelian group. Then there are operations $+: G \times G \rightarrow G$ and $\cdot: \mathbb{Z} \times G \rightarrow G$, which satisfy the following eight properties.

1. For all $a, b, c \in G$, we have $(a + b) + c = a + (b + c)$.
2. There is some $0 \in G$ such that for all $a \in G$, we have $0 + a = a + 0 = a$.

3. For all $a \in G$, there is some $-a \in G$ such that $a + (-a) = -a + a = 0$.
4. For all $a, b \in G$, we have $a + b = b + a$.
5. For all $a \in G$, if $1 \in \mathbb{Z}$ is the number 1, then $1a = a$.
6. For all $n, m \in \mathbb{Z}$ and $a \in G$, we have $(nm)a = n(ma)$.
7. For all $n \in \mathbb{Z}$ and all $a, b \in G$, we have $n(a + b) = na + nb$.
8. For all $n, m \in \mathbb{Z}$ and all $a \in G$, we have $(n + m)a = na + ma$.

Proof. Properties (1)-(4) are simply the axioms of an abelian group (definition 1.1), and thus are automatically true. To prove property (5), note that $1a$ denotes summing over 1 copy of a , and thus $1a = a$. The proofs for property (6)-(8) are longer (but not too tricky), and I only intend to cover the first in the talk. However, I will complete all three in these notes.

(6) Let $n, m \in \mathbb{Z}$ and $a \in G$. We want to show $(nm)a = n(ma)$. Note that

$$(nm)a = \underbrace{a + \cdots + a}_{nm \text{ times}} = \sum_{i=1}^{nm} a.$$

We also have

$$n(ma) = n \left(\sum_{i=1}^m a \right) = \sum_{j=1}^n \sum_{i=1}^m a = \sum_{k=1}^{nm} a.$$

Thus, $n(ma) = (nm)a$.

(7) Let $n \in \mathbb{Z}$ and $a, b \in G$. Then we have

$$n(a + b) = \sum_{i=1}^n a + b.$$

We also have

$$na + nb = \sum_{i=1}^n a + \sum_{i=1}^n b.$$

Since G is abelian, and thus addition is commutative, we can rearrange these sums to

$$\sum_{i=1}^n a + \sum_{i=1}^n b = \sum_{i=1}^n a + b.$$

Thus, $n(a + b) = na + nb$.

(8) Let $n \in \mathbb{Z}$ and $a, b \in G$. We have

$$(n + m)a = \sum_{i=1}^{n+m} a = \sum_{i=1}^n a + \sum_{j=n+1}^{n+m} a = \sum_{i=1}^n a + \sum_{j=1}^m a = na + ma.$$

Thus, we are done. ■

I would like to compare the statements of proposition 1.5 to the definition of a vector space. In linear algebra, you likely learned how to define a vector space where the scalars were the real numbers $\mathbb{R}$. We can also define a *complex* vector space, where the scalars are the complex numbers $\mathbb{C}$. We will work with real vector spaces to keep this discussion simple, but the same discussion applies to any complex vector space. Observe the following.

Definition 1.6. A *vector space* is a set V , together with an operation $+: V \times V \rightarrow V$ called *addition* and an operation $\cdot: \mathbb{R} \times V \rightarrow V$ called *scalar multiplication*, which satisfies the following eight properties.

1. For all $a, b, c \in V$, we have $(a + b) + c = a + (b + c)$.

2. There is some $0 \in V$ such that for all $a \in V$, we have $0 + a = a + 0 = a$.
3. For all $a \in V$, there is some $-a \in V$ such that $a + (-a) = -a + a = 0$.
4. For all $a, b \in V$, we have $a + b = b + a$.
5. For all $a \in V$, if $1 \in \mathbb{R}$ is the number 1, then $1a = a$.
6. For all $n, m \in \mathbb{R}$ and $a \in V$, we have $(nm)a = n(ma)$.
7. For all $n \in \mathbb{R}$ and all $a, b \in V$, we have $n(a + b) = na + nb$.
8. For all $n, m \in \mathbb{R}$ and all $a \in V$, we have $(n + m)a = na + ma$.

I have chosen my notation carefully to make the following observation immediate. Note that if we take definition 3.12, swap out V for G and swap out $\mathbb{R}$ for $\mathbb{Z}$, we get back *exactly* the eight properties of proposition 1.5!

And this leads me to the central topic of this paper. In linear algebra, we required that our scalars come from $\mathbb{R}$, or sometimes $\mathbb{C}$. But *why?* Why not let the scalars be $\mathbb{Z}$, or $\mathbb{Q}$, or polynomials, or smooth functions, or quaternions, or something even more exotic? Why restrict ourselves to $\mathbb{R}$ and $\mathbb{C}$?

The answer is we don't have to. In fact, it's better not to restrict ourselves to $\mathbb{R}$ and $\mathbb{C}$. As we will see, vector spaces (over $\mathbb{R}$ and $\mathbb{C}$) have very nice properties, and are easier to study, hence why they are covered in a linear algebra course. But there are many algebras in reality which are not vector spaces over these two sets of numbers, but over something else. These objects are important in their own right (as we will see many examples), but are not as "well-behaved" as a vector space is. Hence, we use a different name for them: a *module*.

2 Modules

In order to properly introduce modules, we need to be a little more precise about what the scalars actually are. Although we won't be restricting ourselves to $\mathbb{R}$ and $\mathbb{C}$, the scalars still can't be literally *anything*. For example, definition 3.12 property (5) requires the existence of some element 1 in our collection of scalars; property (6) requires that we can multiply scalars with scalars; and property (8) requires that we can add our scalars together. Thus, our collection of scalars has to be some sort of algebra with addition, multiplication, and 1. Such an algebraic structure is referred to by mathematicians as a *ring*.

Definition 2.1. A *ring* is a set R together with *two* operations, addition $+: R \times R \rightarrow R$ and *multiplication* $\cdot: R \times R \rightarrow R$, which satisfy the following properties.

1. The pair $(R, +)$ is an abelian group (definition 1.1).
2. For all $a, b, c \in R$, we have $a(bc) = (ab)c$. This is called *associativity*.
3. There exists some element $1 \in R$ with the property that for all $a \in R$, we have $1a = a1 = a$. This is called *one*, or the *multiplicative identity*, or the *unit*.
4. For all $a, b, c \in R$, we have $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$. These are called the *left and right distributive properties*.

Remark 2.2. Many authors, especially for undergraduate texts, do not require that rings have a unit 1. They would refer to a ring with a unit as a *unital ring*. However, most rings in practice (and all rings I personally work with) have an element 1, so I include a unit in the definition of a ring. This is a fairly standard convention. A ring without a unit is then referred to as an *rng* (we remove the "i" since "i" stands for "identity").

Rings are one of the most general type of interesting algebra that has two operations $+, \cdot$. It's possible to make more general algebras with two operations, but they're much less interesting. For example, without the associative property for multiplication, it would be incredibly hard to work with rings at all. Without the distributive property, there would be no relationship between addition and multiplication, meaning it would be weird to even think of a ring as an algebra with 2 operations.

Example 2.3. The integers $\mathbb{Z}$ are a ring, using normal addition and multiplication. So are the real numbers $\mathbb{R}$, the rational numbers $\mathbb{Q}$, and the complex numbers $\mathbb{C}$. All of these rings have the property that $ab = ba$. Such a ring is known as a *commutative ring*. The natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$ are *not* a ring, since they are not a group. Natural numbers only contain non-negative integers, meaning they don't contain additive inverses, since the inverse of a non-negative integer is negative. However, $\mathbb{N}$ satisfies every other axiom of a ring, yielding a structure known as a *rig* (we remove the “n” since “n” stands for “negative”). I personally know of zero applications of rigs.

Example 2.4. Consider the set of $n \times n$ matrices with real entries, which we will denote by $M_n(\mathbb{R})$. This is a ring using normal matrix addition and multiplication. However, more general $n \times m$ matrices are *not* a ring if $n \neq m$, since we can't multiply two matrices if the dimensions don't line up properly.

Example 2.5. Let R be any *commutative ring*. We can define a new ring called $R[x]$ called the *polynomial ring over R* . Elements of this ring are polynomial functions $R \rightarrow R$, meaning if $p \in R[x]$, then we can write

$$p(x) = \sum_{i=0}^n r_i x^i$$

for some coefficients $r_i \in R$. Addition and multiplication are defined in the exact way you would expect. Polynomial rings are extremely important in *commutative algebra*, which can generally be described as the study of commutative rings (and similar algebras).

We can now define modules over a ring R .

Definition 2.6. Let R be a ring. Then an R -module is an abelian group $(M, +)$ and an operation $\cdot : R \times M \rightarrow M$ called *scalar multiplication* such that the following properties hold:

1. For all $m \in M$, if $1 \in R$ is the unit, then $1m = m$.
2. For all $r, s \in R$ and $m \in M$, we have $(rs)m = r(sm)$.
3. For all $r \in R$ and all $m, n \in M$, we have $r(m + n) = rm + rn$.
4. For all $r, s \in R$ and all $m \in M$, we have $(r + s)m = rm + sm$.

Remark 2.7. This remark should not be given during the talk. There is an alternative definition of a module, which is somewhat more complex but much cleaner. Given rings R, S , a *ring homomorphism* is a function $\varphi : R \rightarrow S$ such that for all $r_1, r_2 \in R$,

$$\varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2) \quad \text{and} \quad \varphi(r_1 r_2) = \varphi(r_1) \varphi(r_2).$$

Let $(M, +)$ be an abelian group, and write $\text{End}_{\text{Ab}}(M)$ for the set of all group homomorphisms $M \rightarrow M$ (we call such a map a *group endomorphism*). If $f, g \in \text{End}_{\text{Ab}}(M)$, then we define the functions $f + g$ and $f \circ g$ as follows: for all $m \in M$,

$$(f + g)(m) = f(m) + g(m), \quad (f \circ g)(m) = f(g(m)).$$

I claim that by taking $+$ to be addition and $\circ$ to be multiplication, the set $\text{End}_{\text{Ab}}(M)$ is a ring. An R -module can then be defined as a ring homomorphism $\rho : R \rightarrow \text{End}_{\text{Ab}}(M)$. Definition 2.6 is then recovered as follows. Let $r \in R$ and $m \in M$; we want to define rm . Since ρ maps R to endomorphisms of M , we have that $\rho(r)$ is a function $M \rightarrow M$. Since $m \in M$, we can evaluate the function $\rho(r)$ at m , yielding an element $\rho(r)(m) \in M$. We define

$$rm := \rho(r)(m).$$

All of the properties listed in definition 2.6 can be proven from this definition.

There are a ton of great examples of R -modules. We will go through a few.

Example 2.8. We have shown previously in proposition 1.5 that any abelian group satisfies all the axioms of a $\mathbb{Z}$ -module. Thus, all abelian groups are $\mathbb{Z}$ -modules. Since all $\mathbb{Z}$ -modules are also abelian groups (condition 1 of definition 2.6), we have a two-sided inclusion

$$\{\mathbb{Z}\text{-modules}\} \subseteq \{\text{abelian groups}\} \quad \text{and} \quad \{\mathbb{Z}\text{-modules}\} \supseteq \{\text{abelian groups}\}.$$

Therefore, $\mathbb{Z}$ -modules are the same thing as abelian groups.

Example 2.9. Let's look at a more concrete example. First, any vector space over $\mathbb{R}$ is an $\mathbb{R}$ -module. Similarly, any vector space over $\mathbb{C}$ is a $\mathbb{C}$ -module. I will later define a *field* as a particular type of ring, and we will define a vector space to be a module over a field.

Example 2.10. A module can be a module over multiple rings at once. For example, any ring R is an abelian group, and thus a $\mathbb{Z}$ -module. However, R is also an R -module, where the scalar multiplication is just normal ring multiplication.

Example 2.11. Here's a more complex, but more concrete and *very useful* example. A *vector field* on $\mathbb{R}^3$ is a function $X : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Recall that $\mathbb{R}^3$ is a vector space, meaning that for any point $(x, y, z) \in \mathbb{R}^3$, the vector field X produces a vector $X(x, y, z) \in \mathbb{R}^3$. Note that if I multiply this vector by a scalar $c \in \mathbb{R}$, I get a new vector $c \cdot X(x, y, z) \in \mathbb{R}^3$. We can use this fact to give the set of vector fields on $\mathbb{R}^3$ the structure of a $C^\infty(\mathbb{R}^3)$ -module.

First, let $\mathfrak{X}(\mathbb{R}^3)$ be the set of all vector fields on $\mathbb{R}^3$. Define $C^\infty(\mathbb{R}^3)$ to be the set of all *smooth functions* $f : \mathbb{R}^3 \rightarrow \mathbb{R}$. Here, smooth simply means that any possible combination of partial derivatives of f exist. We can give $C^\infty(\mathbb{R}^3)$ the structure of a ring by using the ring structure of $\mathbb{R}$. If $f, g \in C^\infty(\mathbb{R}^3)$, then we define the new smooth functions $f + g$ and $f \cdot g$ as follows:

$$\forall (x, y, z) \in \mathbb{R}^3, \quad (f + g)(x, y, z) = f(x, y, z) + g(x, y, z), \quad (f \cdot g)(x, y, z) = f(x, y, z) \cdot g(x, y, z).$$

It's fairly easy to check that this forms a ring. Each ring axiom needs to be checked at a single arbitrary point (x, y, z) of $\mathbb{R}^3$, meaning that we can simply work with equations in $\mathbb{R}$, which is already a ring. Thus, we can consider a $C^\infty(\mathbb{R}^3)$ -module.

In order for $\mathfrak{X}(\mathbb{R}^3)$ to be a $C^\infty(\mathbb{R}^3)$ -module, we need to define a multiplication operation which takes a smooth function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and a vector field $X : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, and outputs a new vector field $fX : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. We define it as follows. Let $(x, y, z) \in \mathbb{R}^3$. then

$$(fX)(x, y, z) = f(x, y, z) \cdot X(x, y, z).$$

Since $f(x, y, z) \in \mathbb{R}$, we have that $f(x, y, z)$ is just a number, i.e. a *scalar*. Since $X(x, y, z)$ is a vector in $\mathbb{R}^3$, we know how to multiply it by a scalar. Thus, multiplying $f(x, y, z)$ and $X(x, y, z)$ can not only be done, but produces a new element of $\mathbb{R}^3$! Thus, $fX : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is indeed a vector field.

Example 2.11 is a more complex example of a module, and it can be difficult to understand at first. However, a *ton* of modules over rings of smooth functions (or other types of functions, eg. continuous, holomorphic) show up in geometry. I won't use many more examples from geometry in these notes to keep them more understandable, but since one of the most interesting applications of module theory is with modules of this form in geometry, it's important (and very cool) to at least know about this.

3 R -linear algebra

A lot of linear algebra still works for R -modules, but not all of it. I will survey some of the main constructions and main results of the theory.

The most striking result in all of module theory is that if R is a random ring, then there is no requirement for an R -module M to have a basis! We can already see a simple example of this by looking at $\mathbb{Z}$ -modules, which we recall (by example 2.8) are simply abelian groups. First, let's recall what a basis is.

Definition 3.1. Let R be a ring, and let M be an R -module. A finite subset $\{m_1, \dots, m_n\} \subseteq M$ is *linearly independent* if the only collection of scalars $r_1, \dots, r_n \in R$ such that

$$r_1 m_1 + \dots + r_n m_n = 0$$

are $r_1, \dots, r_n = 0$. The finite subset $\{m_1, \dots, m_n\} \subseteq M$ is a *spanning set* if for all $x \in M$, there are coefficients $r_1, \dots, r_n \in R$ such that

$$x = r_1 m_1 + \dots + r_n m_n.$$

A *basis* is a set $S \subseteq M$ that is both linearly independent and a spanning set.

Consider the following very simple abelian group. Let $\mathbb{Z}/2$ be the set $\{0, 1\}$ with addition defined by

$$\begin{array}{ll} 0 + 0 = 0 & 0 + 1 = 1 \\ 1 + 0 = 1 & 1 + 1 = 0 \end{array}$$

The only difference from normal addition is $1 + 1 = 0$. We call this the group of *integers modulo 2*. Other notations for this group are $\mathbb{Z}/(2)$, $\mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}_2$. The addition operation is typically written as $n + m \pmod{2}$, but we will drop the $\pmod{2}$.

Since all abelian groups are $\mathbb{Z}$ -modules, we see that $\mathbb{Z}/2$ is a $\mathbb{Z}$ -module. Does this $\mathbb{Z}$ -module have a basis? There are only 4 subsets of $\mathbb{Z}/2$: the empty set, and the sets $\{0\}$, $\{1\}$, $\{0, 1\}$. Clearly the empty set is not spanning. Additionally, $\{0\}$ is not spanning, since there is no way to add 0 to itself in order to get 1. The set $\{1\}$ is spanning, since $2 \cdot 1 = 1 + 1 = 0$. However, it's *not linearly independent*. By the equation we just wrote, $2 \cdot 1 = 0$ for $2 \in \mathbb{Z}$. However, since $2 \neq 0$ in $\mathbb{Z}$, this immediately fails linear independence. The set $\{0, 1\}$ is also not linearly independent for the same reason: $0 \cdot 0 + 2 \cdot 1 = 0$, but $2 \neq 0$.

This immediately makes linear algebra with R -modules much harder. For example, without a basis, we can't define the *dimension* of an R -module. In order to talk about matrices, we need to have a basis; thus we cannot always do matrix algebra in arbitrary R -modules.

Definition 3.2. An R -module M with a basis $S \subseteq M$ is called a *free R -module*. The basis is also referred to as the *generating set* of M .

Although not all R -modules are free (unlike vector spaces), we can still do quite a lot of linear algebra with R -modules. Let's introduce some notions from linear algebra that extend to R -modules without much difficulty.

Definition 3.3. Let R be a ring, and let M, N be R -modules. An *R -linear map* from M to N is a function $\varphi : M \rightarrow N$ with the following properties:

1. For all $m_1, m_2 \in M$, we have $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$.
2. For all $m \in M$ and $r \in R$, we have $\varphi(rm) = r \cdot \varphi(m)$.

Note that definition 3.3 is the same definition as a normal linear map, except the scalars now come from a ring R , rather than $\mathbb{R}$ or $\mathbb{C}$.

Example 3.4. Let M, N be R -modules. The map $0 : M \rightarrow N$ which is defined by $0(m) = 0$ is an R -linear map. We call this the *zero morphism*, the *zero map*, or the *trivial map*. Another example is the map $1_M : M \rightarrow M$ defined by $1_M(m) = m$. This is called the *identity morphism* of M .

Example 3.5. Every linear map between vector spaces you have ever seen is an $\mathbb{R}$ -linear (or $\mathbb{C}$ -linear) map.

We can similarly extend a lot of definitions like vector subspaces, kernels and images to the setting of R -modules.

Definition 3.6. Let R be a ring, and let M, N be R -modules.

1. We say N is a *submodule* of M if $N \subseteq M$ as a set, and if N is closed under addition and scalar multiplication.

2. Let $\varphi : M \rightarrow N$ be an R -linear map. The *kernel* of φ , denoted $\text{Ker } \varphi$, is the submodule

$$\text{Ker } \varphi = \{m \in M \mid \varphi(m) = 0\} \subseteq M.$$

3. Let $\varphi : M \rightarrow N$ be an R -linear map. The *image* of φ , denoted $\text{Im } \varphi$, is the submodule

$$\text{Im } \varphi = \{\varphi(m) \in N \mid m \in M\} \subseteq N.$$

We can also define Cartesian products (which we simply refer to as products), which will be extremely useful.

Definition 3.7. Let R be a ring, and let M and N be R -modules. We define the *product module* $M \times N$ to be the R -module defined as follows.

- The underlying set $M \times N$ consists of all ordered pairs (m, n) , where $m \in M$ and $n \in N$.
- The addition operation is defined *pointwise*, meaning for all $m_1, m_2 \in M$ and $n_1, n_2 \in N$, we have

$$(m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2).$$

- The multiplication operation is also defined pointwise, meaning for all $m \in M$, $n \in N$, and $r \in R$, we define

$$r(m, n) = (rm, rn).$$

Notation 3.8. Due to some complicated but very nice properties of R -modules, the notation $M \times N$ for the product is not used very often. Most of the times, we write $M \times N$ as $M \oplus N$, and call it either the *product* or the *direct sum*. The element (m, n) is often written as $m + n$. This notation makes sense since we can write

$$(m, n) = (m, 0) + (0, n),$$

and then we simply use the notation $m = (m, 0)$ and $n = (0, n)$. This notation turns out to be extremely convenient, and is used extremely often. To avoid confusion, we will not be using this notation (we will be using ordered pairs and $\times$), because this is more aligned with undergraduate linear algebra notation. However, it's important to know that this notation exists.

Remark 3.9. The product of R -modules is both a product *and* a coproduct, in the sense of category theory.

We will now discuss something known as *torsion*, which can make a module fail to have a basis. Let M be an R -module, and let $r \in R$ and $m \in M$ such that $rm = 0$. Suppose that $r \neq 0$. Does $m = 0$? If $R = \mathbb{R}$, then the answer is yes, since we can simply multiply by $1/r$ (since $r \neq 0$) to get

$$0 = \frac{1}{r}rm = \frac{r}{r}m = 1m = m.$$

However, if R is an arbitrary ring, we can't do this. For example, consider the ring of $n \times n$ matrices with real entries $M_n(\mathbb{R})$. Recall that *not all matrices are invertible*. Thus, if $r \in M_n(\mathbb{R})$ is some matrix, there might not be some element r^{-1} allowing us to cancel r in the equation $rm = 0$.

Definition 3.10. Let R be a ring, and M an R -module. An element $m \in M$ is called *torsion* if there is some nonzero $r \in R$ such that $rm = 0$. The *torsion submodule* of M , denoted $\text{Tor}(M)$, is the set

$$\text{Tor}(M) = \{m \in M \mid m \text{ is torsion}\}.$$

It turns out that torsion is the exact obstruction to being free. The only torsion element of a vector space is 0, so if V is a vector space, then $\text{Tor}(V) = \{0\}$. However, there can be R -modules with nonzero torsion elements.

For example, consider the $\mathbb{Z}$ -module $\mathbb{Z}/2$ from before, which we know has no basis. The element 0 is obviously torsion, so let's consider the element 1. Recall that $2 \cdot 1 = 0$, but $2 \neq 0$. Thus, 1 is torsion. We thus have $\mathbb{Z}/2 = \text{Tor}(\mathbb{Z}/2) = \{0, 1\}$.

Note that the submodule $\text{Tor}(M)$ *cannot* be free: given any element $m \in \text{Tor}(M)$, there is some nonzero $r \in R$ with $rm = 0$. Thus, any collection $m_1, \dots, m_n$ of torsion elements admits nonzero scalars $r_1, \dots, r_n \in R$ such that $r_i m_i = 0$ for all i , and thus

$$r_1 m_1 + \dots + r_n m_n = 0.$$

Thus, no nonempty subset of $\text{Tor}(M)$ is linearly independent.

This turns out to be an error with the ring R itself: not all scalars are invertible. Thus, we can look for examples of rings where all (nonzero) elements are invertible.

Definition 3.11. A *field* $\mathbb{K}$ is a ring $(\mathbb{K}, +, \cdot)$ such that the following additional properties hold.

1. $\mathbb{K}$ is a commutative ring, i.e. for all $a, b \in \mathbb{K}$, we have $ab = ba$.
2. For all nonzero $a \in \mathbb{K}$, there is some element $a^{-1} \in \mathbb{K}$ such that $aa^{-1} = a^{-1}a = 1$.

The main three examples of fields are the rational numbers $\mathbb{Q}$, the real numbers $\mathbb{R}$, and the complex numbers $\mathbb{C}$. There are other interesting examples of fields, but we will only consider these three for simplicity.

Definition 3.12. A *vector space* is a module over a field. If $\mathbb{K}$ is a field, a module over $\mathbb{K}$ is called a $\mathbb{K}$ -*vector space*.

The final thing I want to define before we continue past R -modules is the notion of a *quotient*. In higher mathematics, quotients are a ubiquitous construction, but they can be complex to understand at first. The idea is slightly easier to understand in the context of R -modules (and most other algebraic structures) than it is for most objects, like sets or topological spaces. We will begin with some motivation.

Suppose M is an R -module, and R is not a field. It would be nice if M was free, but this would require that M has no torsion, and we cannot know in general if M has no torsion. So we will do the next best thing: given a module M with some nontrivial torsion $\text{Tor}(M) \neq \{0\}$, how do we turn M into a new module, which I will call $M/\text{Tor}(M)$, which has no torsion?

Recall that a torsion element of M is some $m \in M$ such that there is $r \in R$, $r \neq 0$ and $rm = 0$. If this torsion element m is also nonzero, then this is an obstruction to linear independence in M . Thus, we would like the only torsion element to be 0. In other words, if m is a nonzero torsion element in M , we would like to somehow “turn m into 0” when we go to the new module $M/\text{Tor}(M)$. If we can turn all of the torsion elements into 0, then we have gotten rid of the obstructions to linear independence posed by the torsion elements. But how can we possibly construct such a module?

Construction 3.13. Let M be an R -module, and let $N \subseteq M$ be a *submodule*. In the previous motivational section, we took $N = \text{Tor}(M)$, but this construction works more generally, so here we allow N to be any submodule. We will define a new R -module M/N as follows. First, let $m \in M$. We will use the notation $m + N$ to denote the following *set*:

$$m + N := \{m + n \mid n \in N\} \subseteq M.$$

We call $m + N$ a *coset*. We will define M/N to be the collection of all cosets. To define addition, let $m_1, m_2 \in M$. Then we define

$$(m_1 + N) + (m_2 + N) := (m_1 + m_2) + N.$$

Since $m_1 + m_2 \in M$, we have that $(m_1 + m_2) + N$ is a coset. Additionally, since this set consists of all elements $m_1 + m_2 + n$, we can prove that M/N is an abelian group by simply applying the fact that M is an abelian group to the general form of each element of a coset. For example, to prove that addition is commutative, observe that if $m_1 + m_2 + n \in (m_1 + m_2) + N$, then

$$m_1 + m_2 + n = m_2 + m_1 + n \in (m_2 + m_1) + N.$$

Thus, $(m_1 + m_2) + N \subseteq (m_2 + m_1) + N$. The inclusion in the reverse direction is proven in the exact same way.

To define scalar multiplication, if $r \in R$ and $m \in M$, we define

$$r(m + N) := (rm) + N.$$

Since $rm \in M$, we have that $(rm) + N$ is indeed a coset, and we write it as $rm + N$. The rest of the module axioms are proven similarly to the abelian group axioms.

Definition 3.14. Let M be an R -module, and $N \subseteq M$ a submodule. The module M/N defined in ?? 3.13 is called the *quotient* of M by N .

We have now constructed a new module, but that isn't quite good enough. We wanted to *turn M into a new module*. We have that new module now, but we don't have a process for taking an element of m , and turning it into an element of M/N . For this, we need the *natural projection*.

Given any submodule $N \subseteq M$, there is an R -linear map $\pi : M \rightarrow M/N$ called the *natural* or *canonical projection* (also simply referred to as the quotient, or the quotient morphism). This map is defined as follows: if $m \in M$, then

$$\pi(m) = m + N.$$

We can prove that this is R -linear. First, if $m_1, m_2 \in M$, then

$$\pi(m_1) + \pi(m_2) = m_1 + N + m_2 + N = (m_1 + m_2) + N = \pi(m_1 + m_2).$$

Next, if $m \in M$ and $r \in R$, then

$$r\pi(m) = r(m + N) = (rm) + N = \pi(rm).$$

This is the tool we needed for taking elements from M to elements of M/N .

Now let's return to our investigation of $M/\text{Tor}(M)$. Given an element $m \in \text{Tor}(M)$, we want this element to be sent to 0 in the quotient module. The zero element of $M/\text{Tor}(M)$ is the coset $0 + \text{Tor}(M)$. If m is torsion, then we simply need to show that $\pi(m) = m + \text{Tor}(M)$ is equal to $0 + \text{Tor}(M)$.

First, note that $m \in \text{Tor}(M)$. Since $\text{Tor}(M)$ is a submodule of M , it is in particular a module, and thus contains $-m \in \text{Tor}(M)$. Therefore, the element $m + (-m) = 0$ is in the coset $m + \text{Tor}(M)$. Thus, if I have some element $0 + t \in 0 + \text{Tor}(M)$, then I can rewrite

$$0 + t = (m + (-m)) + t = m + (-m + t) \in m + \text{Tor}(M).$$

Here, we note that $-m + t \in \text{Tor}(M)$ since $-m, t \in \text{Tor}(M)$, and since $\text{Tor}(M)$ is a module, it is closed under addition. Thus, $0 + \text{Tor}(M) \subseteq m + \text{Tor}(M)$.

The other inclusion is similar. Let $m + t \in \text{Tor}(M)$. Then since $-m, t \in \text{Tor}(M)$, we have $-m + t \in \text{Tor}(M)$, and thus $m + (-m + t) = 0 + t \in m + \text{Tor}(M)$. Since $0 + t \in 0 + \text{Tor}(M)$, we conclude that $m + \text{Tor}(M) \subseteq 0 + \text{Tor}(M)$. Thus,

$$m + \text{Tor}(M) = 0 + \text{Tor}(M).$$

Finally, note that since $0 \in \text{Tor}(M)$, if an element $\tilde{m} \in M$ is *not* torsion, then $\tilde{m} + 0 = \tilde{m} \in \tilde{m} + \text{Tor}(M)$. Since $\tilde{m} \notin \text{Tor}(M)$, we can conclude that $\tilde{m} + \text{Tor}(M) \neq 0 + \text{Tor}(M)$, and thus $\tilde{m}$ is nonzero. Therefore, the *only* elements of M that are sent to 0 in $M/\text{Tor}(M)$ are the torsion elements. This produces a new module without torsion. It can in fact be shown that this module is free, although that would take more time than we have.

The idea of quotients is very useful. Although they are very abstract constructions, oftentimes they can be easier to understand than some very complex modules (and some special quotients have nicer descriptions than cosets). Thus, being able to reduce a very complex module to a quotient of a more simple module can help us understand it more. I will introduce two extremely powerful theorems of quotients (without proof) to try and illustrate this idea.

Theorem 3.15 (First isomorphism theorem). *Let R be a ring, M, N modules, and $\varphi : M \rightarrow N$ some R -linear map. Then there is an isomorphism $\tilde{\varphi} : M/\text{Ker } \varphi \rightarrow \text{Im } \varphi$, defined by*

$$\tilde{\varphi}(m + \text{Ker } \varphi) := \varphi(m).$$

In particular, if φ is surjective, then $M/\text{Ker } \varphi \cong N$.

Theorem 3.16. *Let V be a finite dimensional vector space over some field $\mathbb{K}$, and let $W \subseteq V$ be a subspace. Then*

$$\dim(V/W) = \dim(V) - \dim(W).$$

4 Chain complexes and homology

We now turn to the usefulness of modules in cooking up invariants. Much of modern mathematics, especially algebraic and geometric fields, is extremely abstract. It can be very difficult to do calculations with these abstract objects, and the solutions to our calculations may themselves be so abstract that it's difficult to extract any meaningful information from them. The modern solution to this issue is *homological algebra*.

Definition 4.1. Let R be a ring. A *chain complex* (of R -modules) $(M_\bullet, d_\bullet)$ is a collection of R -modules $M_\bullet = \{M_i\}_{i \in \mathbb{Z}}$ for each integer $i \in \mathbb{Z}$, and R -linear maps $d_i : M_i \rightarrow M_{i-1}$, which we typically display as the diagram

$$\cdots \xrightarrow{d_{i+2}} M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} \cdots,$$

with the following property: for all i ,

$$d_i \circ d_{i+1} = 0.$$

Here, 0 denotes the zero morphism $0(x) = 0$.

Terminology 4.2. Let $(M_\bullet, d_\bullet)$ be a chain complex. We will often refer to the chain complex simply as $M_\bullet$, and will leave the $d_\bullet$ implicit. The maps d_i are called the *boundary morphisms* or *differentials* due to important examples in geometry. The module M_i is called the *i th degree* of the chain complex M , and an element $m \in M_i$ is said to be an element of *degree i* . We will often write $m \in M_\bullet$ to denote that m is an element of M_i for some $i \in \mathbb{Z}$. We often drop the subscript i on the differential d when it can either be inferred from context, or is not important. We thus often write the requirement that $d_i \circ d_{i+1} = 0$ as the equation $d^2 = 0$, and call this condition “squaring to zero”.

A very obvious question is “why do chain complexes exist?” It's not a crazy thing to imagine a list of vector spaces with maps between them, but why do we require that the maps square to zero? The answer to this question is that we're oftentimes not interested in the chain complex itself. Remember that our goal is to introduce something that is *computable*. Modules are already quite abstract, and so just having a list of modules is not going to simplify anything. However, we can now cook up an invariant of a chain complex, called its *homology*.

Definition 4.3. Let $M_\bullet$ be a chain complex. Notice that since $d_i \circ d_{i+1} = 0$, we must have that for all $x \in \text{Im } d_{i+1}$, it follows that $d_i(x) = 0$. Thus, $x \in \text{Ker } d_i$, meaning

$$\text{Im } d_{i+1} \subseteq \text{Ker } d_i.$$

Note that these are both submodules of M_i . We define the *i th homology* of $M_\bullet$, denoted $H_i(M_\bullet)$, as the module

$$H_i(M_\bullet) = \frac{\text{Ker } d_i}{\text{Im } d_{i+1}}.$$

We define the *homology complex* $H_\bullet(M_\bullet)$ as the chain complex whose i th degree is $H_i(M_\bullet)$, and whose differentials are all the zero morphism.

The reason homology is nice is because it replaces a collection of modules and maps with just a collection of modules. If these modules turn out to be *vector spaces*, then they have a dimension, meaning we are quite literally calculating a single number. It doesn't get much more concrete than that. The honus is then on mathematicians to cook up chain complexes that have *interesting* homology, and luckily we are very good at that. I will give a heuristic sketch of such a complex at the end of the talk.

For now, let's look a little more at why homology is interesting. Homology turns out to be very computable, since a lot of simple properties of chain complexes can immediately tell us things about their homology. For example, let's consider the chain complex

$$\cdots \xrightarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} \cdots,$$

where 0 is the trivial R -module $\{0\}$, the maps $f, g, 0$ are all R -linear, and A, B, C are R -modules. It's a lot of effort to write out the lists of trailing zeros at either end, so we usually only write

$$0 \xrightarrow{0} A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{0} 0.$$

Let's say that A sits in degree 1, B is in degree 0, and C is in degree -1 . Let's say that we know absolutely *nothing* about these modules other than the fact that g is injective and f is surjective. I claim that the only homology that is nonzero is the 0th homology.

Showing that the (-1) th homology is 0 is the easiest. If g is surjective, then $\text{Im } g = C$. Since $0 : C \rightarrow 0$ takes everything to 0, we have $\text{Ker}(0 : C \rightarrow 0) = C$. Thus, $\text{Ker } 0 = \text{Im } g$. As I proved earlier, this implies $\text{Ker } 0 / \text{Im } g = C/C = 0$.

The proof that the 1st homology vanishes requires re-proving a fact from linear algebra. Note that $\text{Im}(0 : 0 \rightarrow A) = \{0\}$, so all I need to prove is that $\text{Ker } f = \{0\}$. We will use the fact that f is injective. Suppose that $m \in \text{Ker } f$. Then $0 = f(m)$. Note that $f(0) = 0$ (exercise: can you prove this?). Thus, $f(m) = f(0)$, and since f is injective, $0 = m$.

The only thing we cannot calculate is the 0th homology $\text{Ker } g / \text{Im } f$. Surjectivity and injectivity are not enough. A sequence of this form (with 3 nonzero modules) is called a *short sequence*. If all of its homologies vanish, then we call it a *short exact sequence*. It turns out that a large portion of homological algebra can be boiled down to the study of short exact sequences via the use of somewhat sophisticated tools. However, these tools are quite tractable (despite their sophistication), and thus we can reasonably reduce many complex problems to this very computable problem.